

I Know You Were Here: Leveraging Probe Request Templates for Identifying Wi-Fi Devices

1st Daniel Vogel
Computer Science IV
University of Bonn
Bonn, Germany
vogel@cs.uni-bonn.de

2nd Felix Viola
Computer Science IV
University of Bonn
Bonn, Germany
viola@uni-bonn.de

3rd Nicholas Malte Kreimeyer
Computer Science IV
University of Bonn
Bonn, Germany
nkreimey@uni-bonn.de

4th Michael Meier
Computer Science IV
University of Bonn, Fraunhofer FKIE
Bonn, Germany
mm@cs.uni-bonn.de

Abstract—Identification of Wi-Fi devices through the observation of sent probe requests (PRs) is commonly used for many different use cases. Successful device identification even when MAC address randomization is used poses a risk to privacy but could also assist law enforcement in re-identifying devices of criminals. In this paper, we examine whether MAC de-randomization techniques provide a sufficiently precise way of re-identifying handheld Wi-Fi devices that have previously been observed. In recent research, MAC address de-randomization techniques are adopted for device counting in various use cases. There, distinguishing devices is utilized without the need to specifically identify individual devices or collecting a maximized amount of data. We propose a system that leverages PR fingerprinting templates which assist the selection of optimized features for fingerprinting to (re-)identify the target device from an observed anonymity group of devices. This system's performance is evaluated on publicly available as well as newly collected data for modern devices and compared to recently proposed MAC address de-randomization techniques. Using a set of experiments, we demonstrate that modern Wi-Fi devices can be perfectly isolated within the anonymity group of received PRs from nearby modern devices for 75 % of the sampled devices.

Index Terms—Data set, WLAN, Wi-Fi, privacy, probe requests, device identification, MAC address, de-randomization, law enforcement, crime prosecution

I. INTRODUCTION

Wi-Fi MAC addresses are long under scrutiny for jeopardizing the privacy of people carrying Wi-Fi devices, as the network discovery process usually entails regular scans through the use of probe requests (PRs), which announce the presence of a device by its MAC address. Since the inception of MAC address randomization as a technique to combat exactly that problem, research show that MAC address randomization has its own slew of problems.

Most importantly, the feature of MAC address randomization is not formally standardized, leading to varying implementations on how MAC addresses are randomized as well as varying device settings on when to randomize which MAC addresses, as discussed in [8]. Older devices may not support MAC address randomization, newer devices may allow to enable or disable it or force its use.

In recent years, the research interest in MAC address de-randomization for the use of device identification rose, as more and more data sets have been published by different research groups. Use cases range from general privacy concerns to

benign device counting and malicious device tracking [1]–[3]. Wi-Fi as a technology is of particular interest, as most people carry wearable Wi-Fi devices, which enable being tracked by them simply carrying the device. The fact that these devices continuously scan for available nearby networks discloses their presence to keen listeners. Installing such a keen listener could help in protecting private properties by identifying intruding persons by their brought devices and collecting data for later re-identification.

In this paper, we propose using PR templates as an assisting method for the process of device identification and MAC de-randomization. Research shows that MAC address randomization is often not enough to protect the users privacy, as within certain anonymity groups devices can potentially be separated from each other through the use of MAC address de-randomization techniques.

There is a need for understanding how individual transmitting devices can be isolated from other nearby transmitting devices but also what information can be gathered and used for re-identification. Re-identification describes the act of using identifying information for a device and using that information to identify the device at a later time in another scenario. The more information and data can be gathered from an observed device, the sounder is the foundation for a later re-identification. All collected information could be used to form tangible leads for law enforcement or in turn attack the privacy of an innocent user in non-crime related scenarios.

This paper answers the following research question: Can a single Wi-Fi device be isolated from all other Wi-Fi devices nearby based on the observed PRs? Our contributions comprise

- an analysis of state of the art PR-based fingerprinting techniques,
- a new data set of PRs captured for devices with MAC address randomization both enabled and disabled,
- an approach utilizing PR templates that can help improve current PR-based fingerprinting techniques.

II. BACKGROUND AND RELATED WORK

By illustrating our use case on a few examples, we lay the ground work for later methodical decisions with regards to data selection and experiment designs.

A. Use case

The use case we work on in this paper is described by the scenario of a Wi-Fi device entering the effective observation range of a listening Wi-Fi device such as a router, which may collect and share observed data from that device. We assume that there is a method to mark the packets received by a device as suspicious, for example through an alert system connected to the Wi-Fi listening system.

The goal is to isolate a target device from other devices by analyzing captured Wi-Fi packets sent by these devices and received by a listening device. The listening device is controlled by the fingerprinter while the sending devices are not. For our analysis we assume the target device to not willingly connect to a Wi-Fi network nearby. Thus we consider target devices to be non-cooperative.

B. Probe Requests

In general, for non-cooperative devices, a connection establishment with a nearby Wi-Fi network is not expected to be observed. Therefore only class 1 frames [4] will be observable, of which PRs are of the most importance to MAC address de-randomization and thus device identification.

PR are widely used in related research, as they carry a lot of information usable for device identification, since this transmitted information is required for network discovery and the preparation of a network connection establishment. Most importantly, PRs contain Wi-Fi MAC addresses for both the sending and receiving device, which are used for device addressing in shared mediums, such as air in the case of Wi-Fi. The seventh bit in the first byte usually denotes whether a MAC address is randomized (if set to 1) or not (if set to 0) [5], which we denote as the *global bit*.

A PR further contains a set of Information Elements (IEs), an example of which is depicted in fig. 1. IEs are used in PRs to signal capabilities and supported features to nearby networks to ease the connection establishment procedure. As there is no formal guideline on how vendors need to implement the usage of IEs, the set of IEs used in PRs across vendors vary drastically, as we show in section V. Since the capabilities and supported features of a particular Wi-Fi device usually do not suddenly change during network discovery, IEs are a valuable asset for device identification procedures. These procedures leverage the heterogeneity of IEs and their values to distinguish between devices.

```

▼ IEEE 802.11 Wireless Management
  ▼ Tagged parameters (30 bytes)
    > Tag: SSID parameter set: Wildcard SSID
    > Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), 6, 9, 12, 18, [Mbit/sec]
    > Tag: Extended Supported Rates 24, 36, 48, 54, [Mbit/sec]
    ▼ Tag: Vendor Specific: Microsoft Corp.: Unknown 8
      Tag Number: Vendor Specific (221)
      Tag length: 7
      OUI: 00:50:f2 (Microsoft Corp.)
      Vendor Specific OUI Type: 8
      Type: Unknown (0x08)
    > Tag: DS Parameter set: Current Channel: 2

```

Fig. 1. An example for IEs in a Wi-Fi PR taken from the UJI data set [6], rendered in wireshark. Shown are a few possible IEs with values listed for the tag: "Vendor Specific".

C. MAC address randomization

Devices that use MAC address randomization most commonly randomize the MAC addresses sent with PRs. Additionally, Fenske et al. postulate that PRs originating from the same device may adjust the contained IEs depending on the MAC address randomization setting chosen [8].

We make use of the *global bit* to decide whether an observed MAC address is randomized or not. This is common practice in related research, as seen in [9]. Though the global bit is meant to indicate whether a MAC address is randomized, not all MAC address implementations abide by that rule, which poses problems, when working with unlabelled data sets of PRs.

D. Related Work

We summarize research on the usage of IEs for device identification, device counting and data sets of PRs with and without MAC randomization.

The use of IEs for device identification and MAC address de-randomization is a fruitful research field. Vanhoef et al. [1] study a fingerprinting approach based on the heterogeneity of IEs to cluster devices based on their PR content rather than on the MAC address. While their work lays the ground work for many years of research to come, they base their investigation on data that is by now over 10 years old and does not account for modern advancements in both the Wi-Fi standard as well as modern devices and their capabilities. We investigate their approach on modern devices and improve on it. They conclude that MAC address randomization does not prevent device identification under their investigation.

Fenske et al. [8] base their work on Vanhoef's idea and investigate their applicability on modern devices. They conclude that modern devices succeed in the usage of MAC address randomization to protect privacy, as some problematic features and IEs like Wi-Fi Protected Setup (WPS) were no longer in wide-spread use. Even though their investigation sheds light on various different aspects, further research shown below indicates that MAC address randomization on modern devices still does not protect against all kinds of analysis.

With recent approaches like [2], [10]–[12] it appears obvious that clustering and machine learning can be a useful tool in separating devices i.e. for the purpose of device counting, though we notice a lack of actual identification evaluation on individual devices. Information on some content and specifically the structure of PRs and IEs gets lost or underrepresented when relying on automated classifiers such as random forest or feature extraction such as Principal Component Analysis (PCA).

Over the years, there were data sets of PRs published with varying degrees of quality and labelling. Bravenec et al. [9] summarize six data sets, of which we chose to use three for our investigation, in addition to our own labelled data set, as described in section IV. Published by Bravenec et al. [6] in 2023, this data set compiles 1.5 million unlabelled PRs, with a large amount of which use MAC address randomization. As this is the most recent data set following a comprehensible

capturing methodology and sporting a decent size, we included this data set. In 2022, Pintor et al. [13] published a data set of labelled measurements over a small amount of devices, which use MAC address randomization and different capturing modes. This is the only data set we found that contains labelled PRs from devices using MAC address randomization, which is important to our investigation, so we included this data set. Even though the Sapienza data set [14] was published in 2013, before the market success of MAC address randomization, this data set has been used in evaluation by multiple researchers, which allows comparison, and for our purpose enables the investigation of the evolution of PR templates. Since it also compiles over 11 million packets, we included this data set.

III. METHODOLOGY

The goal is to isolate a target device from all others using only observed PRs. Towards that, we study how much information is carried by the structure of PRs. We do this by deriving templates and using these to reduce the magnitude of a set of observations, which may allow to work on a higher dimensionality when analyzing the resulting data. An optimized fingerprinting approach leverages these PR templates.

Required are data sets of PRs that allow an analysis of our templating method and the evaluation of the performance of templates as well as a fingerprinting based on these templates. We use PR data sets from different years to analyze the impact of device evolution, where the PRs were sufficiently labelled to have a ground truth to compare to.

Previous work on PR fingerprinting uses various feature extraction techniques like PCA or random forest to pinpoint the IEs with the most overall value towards successfully clustering sets of PRs [2], [10]. Doing this reduces the overall complexity of the data set while maintaining most information, as the IEs with the most overall value to clustering are kept. As noted in section II-B, PRs may contain a plethora of different IEs, with most individual PRs being sparsely populated. This approach trades seemingly marginal information loss for a more manageable calculation complexity and processing overhead.

While some device identification and device counting approaches, as discussed in section II-D, show that reasonable accuracies can be achieved using such feature reduction, anecdotal evidence suggests that uncommon IEs may assist in distinguishing otherwise identical PRs.

During our investigation we observed some fascinating anecdotal evidence that supports the notion of not dropping information for device identification, but rather find ways to improve the performance of identification systems, i.e. through data set compression. Two anecdotal evidence we found are tag 50 `extrates` and the `interworking` bit. `extrates` or `Extended Supported Rates` is used as an extension to `Supported Rates`, which denotes supported rates which do not fit in the limited frame space for the basic supported rates of the sending device. The `interworking` bit denotes the capability of the device to engage in interworking, which is an 802.11u addition which can colloquially be referred to as

“Wi-Fi Roaming” [16]. Tag 50 is not in use as a feature in any published work that uses feature reduction on IEs that we are aware of. Yet, we observed iPhones that sent structurally identical PRs but for the value in tag 50, which allowed us to distinguish them as presented in section V.

TABLE I
EXAMPLES OF DIFFERENT FINGERPRINTS GENERATED FOR SELECTED DEVICES WITH THE METHODOLOGY BY VANHOEF.

Device	Fingerprint
Device1	<code>0,1,50,supra:02040b160c121824,extrates:3048606c</code>
Device2	<code>0,1,50,3,45,127,supra:02040b16,extrates:0c1218243048606c,htcap:002d,htmcsc:000000000000000000000000000000ff,htext:0000,httx:00000000,htasel:00,htampdu:17</code>
Device3	<code>0,1,50,3,221_506f9a_16(020103),supra:02040b16,extrates:0c1218243048606c</code>

TABLE II
EXAMPLES OF DIFFERENT TEMPLATES GENERATED FOR THE SAME SELECTED DEVICES AS IN TABLE I.

Device	Template
Device1	<code>tags;supra;extrate</code>
Device2	<code>tags;supra;extrates;htcap;htmcsc;htext;httx;htasel;htampdu</code>
Device3	<code>tags;221_506f9a_16;supra;extrates</code>

A. Construction and Usage of Probe Request Templates

Our approach uses PR templates to conserve contextual information such as order, amount and existence of all IEs and leverage this information to group PRs by template. This approach results in a meaningful compression of the data set when looking for specific subsets such as all PRs of a particular device. *PR templates* (from here on templates for short) are constructed like follows.

Table III contains any IE we observed within the examined data sets. Introduced tags by Vanhoef et al [1] were used as a starting point and extended by going through the data sets, checking for any new tag, that is not in the list already. Using Wireshark, sensible strings were created to represent the type of information present. Vendor specific tags (221) and extended tags (255) are incorporated generically by using their vendor OUI, vendor type or extended sub-type to form `221_<vendor-OUI>_<type>` and `255_<sub_type>` respectively.

Assuming a target device is known, calculate its template from its PRs. That template can be applied to a set of PRs as a filter template in the following way: For each PR, calculate its template. Compare the calculated template to the filter template. If they match, add the PR to the filtered set. Repeat for all filter templates.

Using this logic, all devices with stable templates (read that generate only one template over all their PRs) have either added all their PRs to the filtered set or none. Devices that do not have stable templates require some additional work. Generally speaking, a device generating more than one template does so, because the IEs or their order change between PRs sent within the observation period. Applying only one of its generated templates will fail to filter all of its sent PRs. We discuss handling of these cases in Section V-C.

TABLE III

LIST OF ALL DISSECTED TAGS IN IES ACROSS ALL EXAMINED DATA SETS WITH THEIR CORRESPONDING VALUE STRINGS USED IN OUR TEMPLATING. VH: VANHOEF, SAP: SAPIENZA, M: MODERN DATA SETS: UJI, CAGLIARI, BONN.

Tag	Descriptive String	VH	Sap	m
	tags	✓	✓	✓
1	supra	✓	✓	✓
3	DS_channel		✓	✓
10	Req_country_info			✓
45	htcap	✓	✓	✓
	htampdu	✓	✓	✓
	htmcs	✓	✓	✓
	htagg		✓	✓
	htext	✓	✓	✓
	httx	✓	✓	✓
	htasel	✓	✓	✓
50	extrates	✓	✓	✓
59	Supported_Operating_Classes_curr			✓
	Supported_Operating_Classes_all			✓
70	RM_En_Cap			✓
107	interworking	✓		✓
	interworking_HESSID			✓
114	MeshID			✓
127	extcap	✓	✓	✓
150	unknown		✓	
191	VHT_Cap_info		✓	✓
	VHT_Sup_MCS		✓	✓
221	221_<vendor-OUI>_<type>	✓	✓	✓
255	255_<sub_type>			✓

To this end, we hypothesize that templates are

- **stable per device**, meaning that all PRs sent by a single device generate the same template,
- **diverse**, meaning that there are many different templates that packets generate,
- contributing to a **significant compression** of the data set, meaning that using a template to filter a data set will filter out a great majority of packets,
- **evolving**, meaning that the evolution of the capabilities of Wi-Fi devices and their features have an impact on the performance of templates.

To test these hypotheses we use four different PR data sets as described in section IV. We aim to gather a greater understanding of how the structure and distribution of IEs can assist in MAC-address-agnostic device identification.

B. Fingerprinting on Filtered Sets

Once the templates have been applied on the data sets, a collection of filtered PRs sets is obtained. Each of these sets only contains PRs matching a single template, in size smaller than the original data set of PRs. We compare the overall performance of a fingerprinting approach on these reduced data sets to previously proposed fingerprinting and clustering approaches.

First, we use the fingerprinting approach proposed by Vanhoef et al. [1], as even though that approach is optimized for data collected before 2016, it might hold up well, especially on the Sapienza data set [14], published in 2013, which we can compare its performance to. We compare the features selected in their paper to our templates and the knowledge base we

gather on IEs over newer data sets to evaluate the evolution of IEs since that publication.

Furthermore, we compare our templates to the features selected in recently published papers that predominantly use clustering on a lower feature count with the goal of mostly device counting.

Finally, we conduct an experiment that models our use case by simulating an intruder that we want to identify. For our use case, an observing Wi-Fi node captures PRs from devices within range over a time span. At a certain time, an intruder breaks into the surveyed perimeter, triggering an alarm. Assuming there is a functioning mechanism that marks a certain PR as *suspicious*, as that PR triggered the alarm, our approach is enabled. We then use our approach to identify all PRs sent by that device through the use of templating, fingerprinting, and optimization of that fingerprinting for each template.

Then a fingerprinting method is selected and used to separate all PRs sent by the devices from all PRs sent by all other devices. For devices, where templating and fingerprinting yielded a flawless isolation from all other devices, no further work is needed. For devices with both a stable template and a stable fingerprint, where there is an overlap with other devices, rendering the isolation flawed, we check, whether the template uses IEs not used as features for the fingerprinting. We add these as features to the fingerprinting and observe the results. For devices with a stable template but unstable fingerprints, we identify mismatches in the fingerprint and check, whether omitting these features from the fingerprinting under that template results in a correct isolation. For devices with an unstable template, we first note the uniqueness of all templates for that device. If all these templates are not shared with other devices, we group all these templates to isolate that device. If these unstable templates are shared with other devices, we try to identify the differences between the fingerprints for those devices and formulate an optimization that way. If finally under a resulting template and resulting fingerprint two or more devices are still not distinguishable, we consider them not to be isolated correctly. In that case, other information such as a modern approach towards sequence numbers or temporal information or Received Signal Strength Indicator (RSSI) could be utilized, which we did not investigate the performance for. This whole process can be automatized. The results are then compared to the ground truth to evaluate the performance of our approach.

IV. DATA SETS

With our methodology stated, next we will describe the data sets we worked on and how they were pre-processed to fit our requirements.

The four data sets used are shown in table IV, where the last one labeled "Bonn" was captured by us. For our investigation, all data sets were pre-processed to acknowledge some requirements. For the unlabelled data sets (**Sapienza** [14], **UJI** [6]), we removed all packets that had their MAC address randomized (see section II-C), as a ground truth is

missing to know for certain, which packets were sent by the same devices. A group of 772 MediaTek MAC addresses making up 31.67% or all devices in that data set, starting with OUI 00-0c-e7, comprise 5271 PRs or 0.56% of all PRs. We concluded that these 772 MAC addresses, while not using the global bit, where all randomized MAC addresses used by the same device, based on PR consistency and temporal information, as no MAC address was reused, once a new MediaTek MAC address was rolled and no two MediaTek MAC addresses sent during overlapping time periods. As such we deemed these PRs as having been sent by the same device for labelling. We then used the non-randomized MAC addresses for labelling PRs as being sent from the device having that MAC address.

Since both of these data sets do not contain reliable ground truths for PRs with randomized MAC addresses, two additional data sets are used, both in much smaller scale, due to the complexity of the capturing and labelling process required to compile such a data set.

The recently published **Cagliari data set** [13] compiles PRs with randomized MAC addresses labelled per device and capturing mode. For this data set, we used the labels provided by the data set to split the data by device and mode. Modes labelled by the data set comprise "screen active/inactive", "Wi-Fi on/off", and "Power saving on/off". We then grouped the PRs for each device/mode for all Wi-Fi channels, as we do not consider channels.

When examining this data set we noticed a lot of problems, that may impact our results. Several device traces were captured in an anechoic chamber, while others were not, none in both. Having a data set captured in both noisy and noise-free scenarios provide a huge opportunity to evaluate, whether our templating approach is robust to these conditions. Additionally, not all devices in the data set are measured in all modes. We chose to include all traces from both scenarios for our evaluation, as we expect these scenarios to have little impact on our approach. Also we evaluated the data set both for each mode and in its entirety, which we call *merged mode*.

As the Cagliari data set [13] does not hold up well to in-depth scrutiny, we decided to compile a **Bonn PR data set** with randomized MAC addresses labelled per device. In particular, we found the claim that devices change the content of their PRs when the MAC randomization mode is changed to not be investigated yet. The data set will be published and steadily filled with more measurements¹.

Our used Wi-Fi capturing system consists of a single Raspberry Pi 4B which is connected to 3 ESP32-S3-DevKit supporting 802.11n Wi-Fi. These ESP32s capture Wi-Fi PRs from the measured devices on the channels 1, 6 and 11 respectively. They transmit the captured frames to the Raspberry Pi via USB-Serial. The Raspberry Pi writes all received Wi-Fi frames into a pcap file. The ESP32s are placed on top of the Raspberry Pi's casing. During the measurement each device is

¹Probe request data set will be updated over time and is available at <https://doi.org/10.60507/FK2/TXOWTX>.

TABLE IV
ANALYSIS OF BOTH DATA SETS AS WELL AS AVERAGE TEMPLATE PERFORMANCE ON THOSE DATA SETS.

data set	Sapienza	UJI	Cagliari	Bonn
#packets	3,943,700	918,054	69,701	4,823
year	2013	2023	2022	2024
#devices	305	1667	22	6
#templates	20	145	25	5
avg. t/d	1.0	1.08	1.68	1.17
avg. d/t	15.25	12.39	1.48	1.4
avg. compression	95.0 %	99.0 %	96.0 %	80.0 %

placed on top of the ESP32, allowing RSSI-based labelling. The devices were measured in a noisy settings with 3 different configurations:

- 1) screen unlocked, Wi-Fi settings activity, Wi-Fi connected, mac randomization disabled, power saving features disabled
- 2) screen locked, Wi-Fi disconnected, mac randomization enabled, power saving features disabled
- 3) screen locked, Wi-Fi disconnected, mac randomization enabled, power saving features enabled

We used the following devices: an iPhone 11, iPhone 13, Redmi 9, Moto g23, Moto g5s, and a Samsung a51.

This pre-processing results in four data sets of PRs, where the labelling allows us to verify whether our device identification approach groups PRs and devices correctly together.

V. EXPERIMENTAL RESULTS

First, we present and discuss in-depth template statistics we observed. Secondly, we summarize the IEs used as features for fingerprinting or clustering respectively as presented in related research with the ones we observed in all data sets. Then the results for our experiments are presented.

A. Template statistics

Using these data sets, we can calculate the template for each PR, as explained above. Having calculated all templates, we can measure metrics that describe the performance of our templating approach on the data sets.

Table IV shows the data sets, amount of PRs after pre-processing per data set, publication year, number of individual Wi-Fi devices based on labelling, total number of unique templates calculated over all PRs in the data set, the average amount of templates that all packets of a single device generated, the average amount of devices that sent PRs that share a template with other devices, and the average amount of data set compression achieved, when applying a template as filter on the whole data set.

From the average statistics in table IV we can infer the following with regards to our previously established hypotheses: There are diverse templates, increasingly diverse as PRs sent by modern devices are used. As more and more features are supported by Wi-Fi devices, different IEs are being used, which directly impacts the templates that can be derived from these PRs.

TABLE V

AMOUNT OF TEMPLATES GENERATED PER DEVICE, BOTH ABSOLUTE AND RELATIVE TO DATA SET SIZE. FOR CAGLIARI AND BONN DATA SETS, THE SHOWN AMOUNTS REPRESENT THE RESULTS FOR THE MERGED MODE. FOR SAPIENZA ALL DEVICES GENERATE ONE TEMPLATE.

#templates per device	UJI		Cagliari		Bonn	
1	1565	94 %	10	45 %	5	83 %
2	81	5 %	9	41 %	1	17 %
3	15	1 %	3	14 %	-	-
4	4	<1 %	-	-	-	-
5	2	<1 %	-	-	-	-

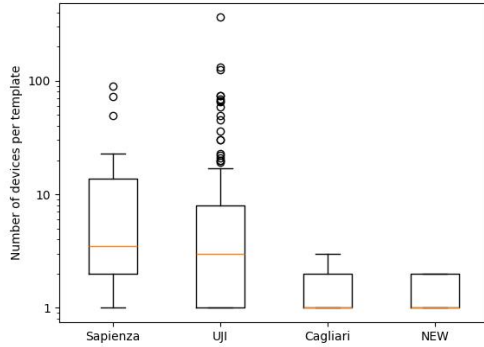


Fig. 2. Amount of devices per template for each data set. The less devices under a template, the better its compression performance when applied as a filter.

For the Sapienza data set, no individual device generated more than one template, meaning that the templates are perfectly stable for that data set. We can see that for all modern data sets, the average amount of templates generated per device is still close to 1.0, but some devices generated more than one template, as can be seen on table V. From data sets like Cagliari and ours we know that the mode that the observed device is in can have an effect on the content of its PRs and therefore the corresponding templates. However, some devices generate multiple templates even though its mode is not changing. It is not clear, why that is. What drives these devices to sometimes seemingly randomly appending IEs that were not used for the last set of probing sweeps? Perhaps the surrounding Wi-Fi environment, nearby networks, a busy medium or something else does impact that behaviour. Since PR data sets do not capture the context in which these PRs have been captured, we simply cannot say for certain. Fig. 2 shows a major difference on how templates are shared between devices between data sets.

B. Comparison of Information Element Usage

The IEs used as features for the approaches Pintor et al. [10] and Uras et al [2] are now compared to the IEs we dissected and presented in table III. When comparing the tag numbers, we observe that most tags are in fact considered by all

approaches. Tags not shared between all approaches are tags 10, 50, 59, 70, 114, and 150. Even though the related research does not use all IEs, they use other additional information like sequence numbers or temporal information. Of all our dissected IEs, the fingerprinting by Vanhoef et al. [1] does not use several for their investigation on the Sapienza data set, as seen in table III. When comparing Sapienza with the modern data sets, several IEs were not yet in use, such as tag 191 VHT, which does make sense, considering that VHT capabilities saw wide spread support after 2013 [15].

C. Results for the Fingerprinting Experiments

As a fingerprinting approach, we applied the fingerprinting proposed by Vanhoef et al. [1] to separate PRs within the target set to single out all PRs sent by only the target device. We omit their proposed sequence number analysis, as modern devices tend to randomize sequence numbers. We reference that fingerprinting approach as *naive fingerprinting*. Then, we extracted knowledge from templates and fingerprints to find optimized fingerprinting features for each template, which we then applied again to observe potential performance improvements.

Our results are summarized in table VI. Only if all PRs sent by a device are correctly grouped towards that sending device, we consider the fingerprinting accurate. Our results show that through some careful investigation of the templates and fingerprints 56 % of all modern devices were matched with exactly their sent PRs before and 75 % after optimizing the fingerprinting based on the gathered templating knowledge. The optimization in fact achieves a drastic performance improvement of the fingerprinting for that particular observation of PRs. These results point towards the value that a large knowledge base could bring to device identification systems, as the application of expert knowledge for PR templates promises a meaningful performance improvement. Devices that were not able to be isolated were those, that had fingerprinting after optimization that overlapped with fingerprints from other devices. We noticed that e.g. for the Cagliari data set B, E and J overlapped, all of whom are Xiaomi phones as well as L and V overlapped, which were Google Pixel phones. The devices in our Bonn data set that we could not isolate were the same devices that the naive approach could not isolate, so our optimization did not yield any improvement on that data set.

A comparison with the results of other research e.g. on the Cagliari data set, we need to account for the difference in use cases. Pintor et al. [10] use clustering methods and compare their amount of clusters with the actual device count. When comparing our count of optimized template/fingerprint groups on the Cagliari data set mode S, we count 20 groups for 22 devices. Though our approach does not aim for an accurate device count but rather for a perfect isolation of devices, at least for that tested set it can be stated, that our approach would be a viable alternative.

TABLE VI

RESULTS FOR THE TEMPLATING EXPERIMENTS. SHOWN ARE THE PERCENTAGE OF ALL DEVICES THAT HAVE ALL THEIR PRs ISOLATED PERFECTLY WHEN APPLYING NAIVE FINGERPRINTING AND AFTER OPTIMIZING THE FINGERPRINTING USING EXPERT KNOWLEDGE.

data set	Sapienza	UJI	Cagliari	Bonn
isolation naive FP	40 %	56 %	54.5 %	66.6 %
isolation optimized FP	40 %	75 %	77.3 %	66.6 %

VI. DISCUSSION AND FUTURE WORK

The observation of large, generalized templates was something we expected, as one could expect to see a certain "best practice" for the construction of PRs, leading to many devices using the same IEs for their PRs. The magnitude of the outlier templates are surprising, however, and so is that the templates are not particularly "generalized" as they actually are most often generated by the same family of devices, instead of being generated by i.e. many android phones of different brands. This observation could speak for a certain imbalance within the data set, but it could also speak for certain biases, when capturing PRs in the wild. For example, since the UJI data set was captured in Spain, in part during a holiday week, the common market distribution of owned Wi-Fi devices by people in Spain may affect the overall transferability of our UJI observations to other data sets.

The same is true when comparing the capturing modes between different data sets. Between UJI, Cagliari and Bonn, these data sets have one common measurement mode, which only a small partition of the Cagliari and Bonn data set was measured in. Obviously these data sets were compiled with different goals in mind, thus diverging in experimental design and measurement setup.

Finally, our newly published data set is still rather small and it is questionable, whether the relatively small sample size allows to conclude the impact of the MAC randomization mode on the PRs and their generated templates. Yet, there is no other labelled data set that would allow such an investigation to the best of our knowledge. Building a knowledge base as an asset to assist fingerprinting approaches through the usage of templating may be required in order to evaluate whether these expert knowledge-based optimizations hold up for generalized use cases, especially those with many more simultaneously observed devices.

We plan to populate our Bonn data set with many more devices to increase the validity of any investigation relying on it. For this, we are in contact with an institution that houses thousands of Wi-Fi devices that we may have the opportunity to compile measurements from. This would allow the construction of a larger knowledge base and a more comprehensive evaluation.

Another yet missing investigation is how these device identifying data collections perform in practice, when utilized for actual device re-identification e.g. in the use case of criminal investigations.

VII. CONCLUSION

In this paper, we investigate the performance of a Wi-Fi PR templating approach when applying it to identify all PRs sent by a target device. We describe the construction and usage of Wi-Fi PR templates and how to evaluate their quality. This templating approach is then applied to four PR data sets in order to measure the amount of templates generated by single devices, the amount of devices that share templates and the compression rate that the filtering using a template has on the whole data set.

We describe the evolution of this templating approach by applying it to a data set over 10 years old by comparing the differences and find that the amount of IEs contributing to the generated templates increased and thus a more diverse set of templates helps to differentiate modern devices better by 48 percent points. A series of experiments is conducted to evaluate the impact of the templating approach on device identification. The approach is applied to a set of PRs captured from a set of devices, and is then followed up by a fingerprinting optimization approach in order to identify all PR captured from a single target device. We found our approach to result on average in 75 % of devices being correctly isolated. These information could be used by law enforcement to re-identify devices that have been observed at crime scenes in order to strengthen home security.

VIII. ACKNOWLEDGMENTS

The work on this paper has been funded by the German Federal Ministry of Education and Research (BMBF) for the project WACHMANN (WLAN-based recording of mobile device characteristics for alarming and fighting property crime) under grant number 13N15357.

REFERENCES

- [1] Vanhoef, M., Matte, C., Cunche, M., Cardoso, L. S., & Piessens, F. (2016, May). Why MAC address randomization is not enough: An analysis of Wi-Fi network discovery mechanisms. In *Proceedings of the 11th ACM on Asia conference on computer and communications security* (pp. 413-424).
- [2] Uras, M., Ferrara, E., Cossu, R., Liotta, A., & Atzori, L. (2022). MAC address de-randomization for WiFi device counting: Combining temporal-and content-based fingerprints. *Computer Networks*, 218, 109393.
- [3] Yu, L., Luo, B., Ma, J., Zhou, Z., & Liu, Q. (2020). You are what you broadcast: Identification of mobile and IoT devices from (public)Wi-Fi. In *29th USENIX security symposium (USENIX security 20)* (pp. 55-72).
- [4] IEEE Computer Society LAN/MAN Standards Committee and others (2007). IEEE Standard for Information technology-Telecommunications and information exchange between systems-Local and metropolitan area networks-Specific requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications. IEEE Std 802.11.
- [5] Eastlake 3rd, D., & Abley, J. (2013). IANA Considerations and IETF Protocol and Documentation Usage for IEEE 802 Parameters (No. rfc7042).
- [6] Bravenec, T., Torres-Sospedra, J., Gould, M., & Fryza, T. (2023, September). UJI probes: Dataset of Wi-Fi probe requests. In *2023 13th International Conference on Indoor Positioning and Indoor Navigation (IPIN)* (pp. 1-6). IEEE.
- [7] Chatzisofroniou, G., & Kotzanikolaou, P. (2021). Association attacks in ieee 802.11: Exploiting wifi usability features. In *Socio-Technical Aspects in Security and Trust: 9th International Workshop, STAST 2019, Luxembourg City, Luxembourg, September 26, 2019, Revised Selected Papers 9* (pp. 107-123). Springer International Publishing.

- [8] Fenske, E., Brown, D., Martin, J., Mayberry, T., Ryan, P., & Rye, E. (2021). Three years later: A study of mac address randomization in mobile devices and when it succeeds. *Proceedings on Privacy Enhancing Technologies*.
- [9] Bravenec, T., Torres-Sospedra, J., Gould, M., & Fryza, T. (2023). UJI Probes Revisited: Deeper Dive Into the Dataset of Wi-Fi Probe Requests. *IEEE Journal of Indoor and Seamless Positioning and Navigation*.
- [10] Pintor, L., & Atzori, L. (2022, December). Analysis of Wi-Fi Probe Requests Towards Information Element Fingerprinting. In *GLOBECOM 2022-2022 IEEE Global Communications Conference* (pp. 3857-3862). IEEE.
- [11] Hao, L., Huang, B., Jia, B., & Mao, G. (2023). On the Fine-Grained Crowd Analysis via Passive WiFi Sensing. *IEEE Transactions on Mobile Computing*.
- [12] Delzanno, G., Caputo, L., D'Agostino, D., Grosso, D., Mustajab, A. H., Bixio, L., & Rulli, M. (2023). Automatic Passenger Counting on the Edge via Unsupervised Clustering. *Sensors*, 23(11), 5210.
- [13] Pintor, L., & Atzori, L. (2022). A dataset of labelled device Wi-Fi probe requests for MAC address de-randomization. *Computer Networks*, 205, 108783.
- [14] Barbera, M. V., Epasto, A., Mei, A., Kosta, S., Perta, V. C., & Stefa, J. (2013). CRAWDAD dataset sapienza/probe-requests (v. 2013-09-10).
- [15] IEEE Standard for Information technology–Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications–Amendment 4: Enhancements for Very High Throughput for Operation in Bands below 6 GHz. IEEE.
- [16] IEEE Standard for Information technology–Part 11: Wireless LAN medium Access Control (MAC) and Physical Layer (PHY) Specifications–Amendment 9: Interworking with External Networks. 2011.